NATIONAL MATHEMATICS CONTEST

A-LEVEL PAPER 1.

SATURDAY 17TH MARCH, 2018 9:00 AM - 12:15 PM

INSTRUCTIONS TO CANDIDATES AND SUPERVISORS:

(a) *This competition is conducted on the assumption that proper security is maintained. UMS reserves the right, should there be evidence or suspicion of any malpractice, to reject scripts from a particular school/institution.*

(b) *There are* **two** *sections and each section carries* 50 *marks. Attempt all questions in exactly three hours fifteen minutes (*$9:00am$ *to* $12:15pm$*). Marks will be awarded for only answers for which a clear and logical layout of the working is provided.*

(c) *Answer scripts of only qualified students should be returned to the contest coordinator, latest* **Monday 19th March, 2018**.

(d) **ALL** *participants who qualify for Paper* 2 *are cordially invited to the certificate and prize giving ceremony on Saturday* 28*th July,* 2018 *at NOON at Makerere University, Kampala.*

(e) *National Mathematics Contest* 2018 *Paper* 2 *will be done on* 23*rd June,* 2018 *at various centres.*

SECTION A : 5 marks each

1. Let $\dfrac{113}{50} = a + \dfrac{1}{b + \dfrac{1}{c + \dfrac{1}{d + \dfrac{1}{e}}}}$, where a, b, c, d and e are integers. Find c and e.

2. Suppose real numbers x, y, z satisfy the equation $\dfrac{x}{y+z} + \dfrac{y}{z+x} + \dfrac{z}{x+y} = 1$.
   Compute the value of
   $$\frac{x^2}{y+z} + \frac{y^2}{z+x} + \frac{z^2}{x+y}$$

3. Let ABC be a triangle and D be a point such that A and D are on opposite sides of BC. Given that $\angle ACD = 75°, AC = 2, BD = \sqrt{6}$, and AD is an angle bisector of both $\triangle ABC$ and $\triangle BCD$, find the area of quadrilateral $ABDC$.

4. Liesl has a bucket. Henri drills a hole in the bottom of the bucket. Before the hole was drilled, Tap A could fill the bucket in 16 minutes, tap B could fill the bucket in 12 minutes, and tap C could fill the bucket in 8 minutes. A full bucket will completely drain out through the hole in 6 minutes. Liesl starts with the empty bucket with the hole in the bottom and turns on all three taps at the same time. How many minutes will it take until the instant when the bucket is completely full?

5. If $f(x) = a + bx$, what are the real values of a and b such that
   $$\begin{aligned} f(f(f(1))) &= 29 \\ f(f(f(0))) &= 2 \end{aligned}$$

6. Add all the digits in 2008^{2008}. Add all the digits in the resulting number. Keep going until the result has only one digit. What is this digit?

7. Let $x = \log_{10}(81)$ and $y = \log_{10}(25)$. If one expresses $\log_{10}(6)$ as $ax + by + c$ where a, b, and c are rational numbers, then what is $a + b + c$?

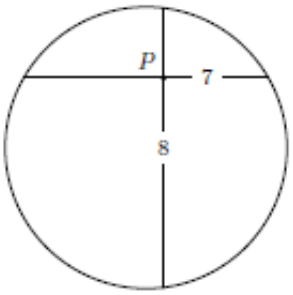
8. Three identical circles of radius 3 lie externally tangent to each other. A fourth, larger circle is drawn around the other three circles so that the smaller three circles are internally tangent to the larger circle. Compute the radius of the larger circle.

9. Ed writes the first 2018 positive integers down in order: $1, 2, 3, \ldots, 2018$. Then for each power of 2 that appears, he crosses out that number as well as the number 1 greater than that power of 2. After he is done, how many numbers are not crossed out?

10. If f is a function such that $f(x-1) = x^2 - 3x$, then $f(x+1) =$?

SECTION B : 10 marks each

11. Two perpendicular chords of a circle intersect at point P. One chord is 7 units long, divided by P into segments of length 3 and 4, while the other chord is divided into segments of length 2 and 6. What is the diameter of the circle?



12. (a) Let $f(x) = \ln\left(\dfrac{1+x}{1-x}\right)$. Determine $f\left(\dfrac{3x+x^3}{1+3x^2}\right)$ in terms of $f(x)$.

    (b) If $x^2 + xy + y^2 = 84$ and $x - \sqrt{xy} + y = 6$, what is xy?

    (c) How many real solutions does the following equation have?

$$x^2 + \sqrt{x-2} = 4 + \sqrt{4-2x}.$$

13. Cindy has a collection of identical rectangular prisms. She stacks them, end to end, to form 1 longer rectangular prism. Suppose that joining 11 of them will form a rectangular prism with 3 times the surface area of an individual rectangular prism. How many will she need to join end to end to form a rectangular prism with 9 times the surface area?

14. If $\dfrac{1}{x} + \dfrac{1}{y} = \dfrac{1}{2}$ and $\dfrac{1}{x+1} + \dfrac{1}{y+1} = \dfrac{3}{8}$, compute $\dfrac{1}{x-1} + \dfrac{1}{y-1}$ as a fraction of integers.

15. (a) Let $a_1, a_2, a_3, a_4, a_5, \ldots$ be a geometric progression with positive ratio such that $a_1 > 1$ and

$$(a_{1357})^3 = a_{34}.$$

    Find the smallest integer n such that $a_n < 1$.

    (b) If $a > 0$ and $b > 0$ are real numbers satisfying

$$\frac{1}{a} + \frac{1}{b} \leq 2\sqrt{2} \quad \text{and} \quad (a-b)^2 = 4(ab)^3,$$

    what is the value of $\log_a b$?

SECTION A : 5 marks each

1. By expansion

$$\begin{aligned}
\frac{113}{50} &= 2+\frac{13}{50}\\
&= 2+\frac{1}{\frac{50}{13}}\\
&= 2+\frac{1}{3+\frac{11}{13}}\\
&= 2+\frac{1}{3+\frac{1}{\frac{13}{11}}}\\
&= 2+\frac{1}{3+\frac{1}{1+\frac{2}{11}}}\\
&= 2+\frac{1}{3+\frac{1}{1+\frac{1}{\frac{11}{2}}}}\\
&= 2+\frac{1}{3+\frac{1}{1+\frac{1}{5+\frac{1}{2}}}}
\end{aligned}$$

Therefore $(a,b,c,d,e)=(2,3,1,5,2)$. Therefore $(c,e)=(1,2)$

2. We first note that $x+y+z \neq 0$. Otherwise,

$$\frac{x}{y+z}+\frac{y}{z+x}+\frac{z}{x+y}=-3$$

Thus, we have

$$\left(\frac{x}{y+z}+\frac{y}{z+x}+\frac{z}{x+y}\right)(x+y+z)=(x+y+z).$$

Simplifying it, we have

$$\frac{x^2}{y+z}+x+\frac{y^2}{z+x}+y+\frac{z^2}{x+y}+z=(x+y+z),$$

which yields

$$\frac{x^2}{y+z}+\frac{y^2}{z+x}+\frac{z^2}{x+y}=0$$

3. Since AD is an angle bisector of both ABC and BCD, $\angle BAD = \angle CAD$ and $\angle BDA = \angle CDA$. Then by angle-side-angle congruence, $\triangle ABD \cong \triangle ACD$, and $CD = BD = \sqrt{6}$. Since $\angle ACD = 75°$, we can use the area formula

$$[ACD] = \frac{1}{2} AC \cdot CD \cdot \sin 75° = \frac{1}{2} \cdot 2 \cdot \sqrt{6} \cdot \frac{\sqrt{2}+\sqrt{6}}{4} = \frac{\sqrt{3}+3}{2}$$

Because $[ABD] = [ACD]$ we have $[ABCD] = 2[ACD] = 3 + \sqrt{3}$.

Solution 2: We begin as in the original solution by noticing $\triangle ABD \cong \triangle ACD$. This implies that $ABCD$ is a kite, which means that $AD \perp BC$. Let E be the intersection of AD and BC. Note that then $\triangle CAE$ is a $30-60-90$ right triangle, while $\triangle CDE$ is a $45-45-90$ right triangle. This gives us $AD = 1 + \sqrt{3}$ and $BC = 2\sqrt{3}$, so the area of kite $ABCD$ is $\frac{1}{2} AD \cdot BC = 3 + \sqrt{3}$

4. Since taps A, B and C can fill the bucket in 16 minutes, 12 minutes and 8 minutes, respectively, then in 1 minute taps A, B and C fill $\frac{1}{16}$, $\frac{1}{12}$ and $\frac{1}{8}$ of the bucket, respectively.

   Similarly, in 1 minute, $\frac{1}{6}$ of the bucket drains out through the hole. Therefore, in 1 minute with the three taps on and the hole open, the fraction of the bucket that fills is

$$\frac{1}{16}+\frac{1}{12}+\frac{1}{8}-\frac{1}{6}=\frac{3+4+6-8}{48}=\frac{5}{48}.$$

   Thus, to fill the bucket under these conditions will take $\frac{48}{5}$ minutes.

5. We have that

$$\begin{aligned} f(f(f(1))) &= f(f(a+b)) = f(a+ab+b^2)) = a+ab+ab^2+b^3 = 29 \\ f(f(f(0))) &= f(f(a)) = f(a+ab)) = a+ab+ab^2 = 2 \end{aligned}$$

Therefore $2+b^3 = 29 \Leftrightarrow b = 3$ and $a = \dfrac{2}{13}$.

6. You can recognise the pattern

n	2008^n	Sum of digits
0	1	1
1	2008	1
2	4,032,064	1
3	8,096,384,512	1
4	16,257,540,100,096	1

Or, we know that, the sum of digits

$$\begin{aligned} 2008^{2008} &= (2008 \mod 9)^{2008} \\ &= 1^{2008} \\ &= 1 \end{aligned}$$

7. Solution: Since $81 = 3^4$, we get $\log_{10}(3) = \dfrac{x}{4}$. Also, $25 = 5^2$, so we obtain $\log_{10}(5) = \dfrac{y}{2}$.

Note that $\log_{10}(5) + \log_{10}(2) = 1$, so $\log_{10}(2) = 1 - \dfrac{y}{2}$. Thus,

$$\log_{10}(6) = \log_{10}(2) + \log_{10}(3) = 1 - \frac{y}{2} + \frac{x}{4}.$$

Therefore $a = \dfrac{1}{4},\ b = -\dfrac{1}{2}$ and $c = 1$, such that $a + b + c = \dfrac{1}{4} - \dfrac{1}{2} = 1 = \dfrac{3}{4}$

8. Three identical circles of radius 3 lie externally tangent to each other. A fourth, larger circle is drawn around the other three circles so that the smaller three circles are internally tangent to the larger circle. Compute the radius of the larger circle.

Solution 1 *Let A, B, C be the centers of the three smaller circles, and let O be the center of the larger circle. Note that the radius of the larger circle is simply $3 + OA$. Because the three smaller circles are tangent to each other, we have $AB = AC = BC = 3 \cdot 2 = 6$, so $\triangle ABC$ is an equilateral triangle. Now if we let D be the midpoint of AB, we can see that ADO is a $30-60-90$ triangle. This gives us the ratio $\frac{\sqrt{3}}{2} = \frac{3}{OA}$, which we solve to get $OA = 2\sqrt{3}$. Therefore, the radius of the larger circle is $3 + 2\sqrt{3}$.*

9. Ed writes the first 2018 positive integers down in order: $1, 2, 3, \ldots, 2018$. Then for each power of 2 that appears, he crosses out that number as well as the number 1 greater than that power of 2. After he is done, how many numbers are not crossed out?

   There are 11 powers of 2 less than 2018:

   $$2^0, 2^1, \ldots, 2^{10} = 1024.$$

   For each power of 2, we cross out two numbers. However, we end up crossing out 2 twice, because it is crossed out by $2^0 + 1$ and by 2^1. Therefore, the total number of integers we cross out is $11 \cdot 2 - 1 = 21$. It follows that $2018 - 21 = 1997$ numbers are left.

10. We know $f(x-1) = x^2 - 3x$. Let $y = x - 1 \quad \Leftrightarrow \quad x = y + 1$. Then

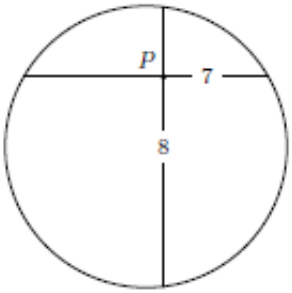
$$\begin{aligned} f(y) &= f(x-1) = x^2 - 3x \\ &= (y+1)^2 - 3(y+1) \\ &= y^2 + 2y + 1 - 3y - 3 \\ f(y) &= y^2 - y - 2 \end{aligned}$$

or $f(x) = x^2 - x - 2 \quad \Leftrightarrow$

$$\begin{aligned} f(x+1) &= (x+1)^2 - (x+1) - 2 \\ &= x^2 + x - 2 \end{aligned}$$

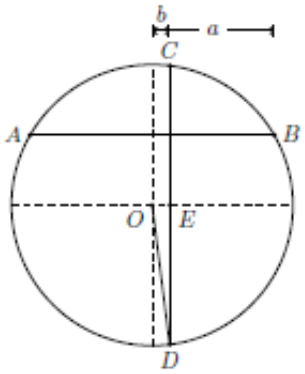
SECTION B : 10 marks each

For each question in SECTION B, your solution must be well-organized and contain words of explanation or justification. Marks are awarded for completeness, clarity, and style of presentation. A correct solution, poorly presented, will not earn full marks.



11. Two perpendicular chords of a circle intersect at point P. One chord is 7 units long, divided by P into segments of length 3 and 4, while the other chord is divided into segments of length 2 and 6. What is the diameter of the circle?

Solution 2 *Label points A, B, C, D, E and the center of the circle, O, in the diagram below (right), draw diameters parallel to both chords, and assign distances a, b as depicted. Since $\overline{CD}$ is the longer*



chord of length 8, the distance DE must equal 4. Since $\overline{AB}$ is the shorter chord of length 7, it must be that $a = 3$ and $a + b = 3.5$. This gives $b = 0.5$. By the Pythagorean Theorem, it must be that $r = OD = \dfrac{\sqrt{65}}{2}$.

12. (a)

$$\begin{aligned} f\left(\frac{3x+x^3}{1+3x^2}\right) &= \ln\left(\frac{1+\frac{3x+x^3}{1+3x^2}}{1-\frac{3x+x^3}{1+3x^2}}\right) \\ &= \ln\left(\frac{1+3x^2+3x+x^3}{1+3x^2-3x-x^3}\right) \\ &= \ln\left(\frac{(1+x)^3}{(1-x)^3}\right) \\ &= 3f(x) \end{aligned}$$

(b) Rewrite the second equation as $x+y=6+\sqrt{xy}$. Squaring then gives

$$x^2+y^2+2xy = 36+xy+12\sqrt{xy}.$$

Subtracting this equation from $x^2+xy+y^2=84$ and simplifying gives $48=12\sqrt{xy}$, thus $xy=16$.

(c) The square roots are defined only when $x-2\geq 0$ and $4-2x\geq 0$. The two inequalities are satisfied only when $x=2$. Since $x=2$ is a solution, the equation has exactly one real solution.

13. Let n be the number of boxes stacked together, and let m be the resulting multiplier on the surface area (i.e. the resulting box has m times the surface area of an individual box). Letting x, y be the lengths of the sides of the box and z be the height of the box, we may write

$$\begin{aligned} 2xy + 2x(nz) + 2y(nz) &= m(2xy + 2xz + 2yz) \\ (n-m)(x+y)z &= (m-1)xy \\ \frac{n-m}{m-1} &= \frac{xy}{(x+y)z} \end{aligned}$$

Note that the right side is a constant because x, y, z are fixed. Furthermore, we are given $n = 11$ and $m = 3$, so plugging this in gives us $\frac{11-3}{3-1} = 4 = \frac{xy}{(x+y)z}$. Therefore, we must find n such that $\frac{n-9}{9-1} = 4$, which solves to $n = 41$.

Solution 2: Note that after stacking 10 additional boxes, you gain 2 additional boxes worth of surface area. This ratio is constant, so if we need n additional boxes to get 8 additional boxes worth of surface area, we have the equation $\frac{n}{8} = \frac{10}{2}$. Solving yields $n = 40$, which means we need a total of 41 boxes.

14. If $\frac{1}{x}+\frac{1}{y}=\frac{1}{2}$ and $\frac{1}{x+1}+\frac{1}{y+1}=\frac{3}{8}$, compute $\frac{1}{x-1}+\frac{1}{y-1}$.

Solution 3 *With two equations and two unknowns, we can solve for the expressions $a=xy$ and $b=x+y$. The first equation can be written as*

$$\frac{1}{x}+\frac{1}{y}=\frac{x+y}{xy}=\frac{b}{a}=\frac{1}{2}$$

which implies that $a=2b$. Similarly, from the second equation we have

$$\frac{1}{x+1}+\frac{1}{y+1}=\frac{(x+y)+2}{xy+(x+y)+1}=\frac{b+2}{a+b+1}=\frac{3}{8}$$

which implies that $8(b+2)=3(a+b+1)$, or $3a-5b=13$. Solving the system of equations gives us $a=26$ and $b=13$. Therefore, we have

$$\frac{1}{x-1}+\frac{1}{y-1}=\frac{(x+y)-2}{xy-(x+y)+1}=\frac{b-2}{a-b+1}=\frac{11}{14}$$

15. (a) Let $a_1, a_2, a_3, a_4, a_5, \ldots$ be a geometric progression with positive ratio such that $a_1 > 1$ and $(a_{1357})^3 = a_{34}$. Find the smallest integer n such that $a_n < 1$.

Let r be the ratio between the terms. The n-th term of the sequence can therefore be written as $a_n = a_1 r^{n-1}$. This allows us to write

$$(a_{1357})^3 = a_{34} \;\Rightarrow\; \left(a_1 r^{1356}\right)^3 = a_1 r^{33} \;\Rightarrow\; a_1^2 r^{4035} = 1.$$

Since $a_1 > 1$, we must have $0 < r < 1$, so the geometric progression is decreasing. However, the above equation also gives

$$a_1^2 r^{4035} = \left(a_1 r^{2017}\right)\left(a_1 r^{2018}\right) = a_{2018} a_{2019} = 1.$$

Because the sequence is decreasing, we must have $a_{2018} > 1 > a_{2019}$, which gives us the answer 2019.

(b) From $\dfrac{1}{a} + \dfrac{1}{b} \leq 2\sqrt{2}$, we have $0 < a + b \leq 2\sqrt{2}ab$, then $0 < (a+b)^2 \leq 8(ab)^2$. So we get

$$4(ab)^3 = (a-b)^2 = (a+b)^2 - 4ab \leq 8(ab)^2 - 4ab$$

Now let us focus on $4(ab)^3 \leq 8(ab)^2 - 4ab$. Since $ab > 0$, we may divide both sides of previous inequality by $4ab$ and rearrange the inequality, giving $(ab)^2 - 2ab + 1 \leq 0$ and then $(ab-1)^2 \leq 0$, which can only be true if $ab = 1$. So $b = a^{-1}$, i.e., $\log_a b = -1$.